MIL-H-15289B(SHIPS)
10 January 1963
SUPERSEDING
MIL-H-15289A(SHIPS)
28 July 1959

MILITARY SPECIFICATION

HEATERS, SPACE, ELECTRIC, RADIANT, PANEL TYPE

1. SCOPE

1.1 This specification covers radiant air heaters, electric panel type, intended for improving habitability conditions in the living and working spaces onboard Naval ships. The heaters are used for isolated spaces requiring radiant type heat over extended surfaces.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 350 5th Avenue, New York 1, N.Y.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N.Y.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General characteristics.- Radiant heaters consist of a low temperature panel, suitable for permanent bulkhead or overhead mounting. Heaters shall be mounted by means of flanges provided on the edges of the panel.

FSC 4520

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3.2 Size and wattage rating.- Panels shall be rectangular in shape and as far as is practicable shall be constructed for a uniform thickness not greater than 7/8 inch. The thickness of the edge of the panel shall not exceed 7/8 inch but because of the tendency of the panels to bow, thickness as great as 1-3/8 inches near the center of the panel are acceptable. Unless otherwise specified in the contract or order, heaters shall be furnished in the standard size and wattage specified in 3.2.1. Heaters may be ordered in special sizes required for specific applications (see 6.1).

3.2.1 Standard size and wattage.- The standard size heater shall conform to the dimensions shown on figure 1. The wattage shall be 900 watts ± 10 percent.

3.2.2 Special sizes.- The overall length and width of the heater (including mounting flanges) and wattage (± 10 percent) shall be as specified (see 6.1). The number and spacing of mounting bolt holes may be specified, if desired (see 6.1). If not specified, the number and spacing of bolt holes shall be selected by the manufacturer. The wattage shall not exceed 150 watts per square foot of heated panel area. Flock shall be used on all panels where the wattage exceeds 75 watts per square foot of heated panel area.

3.3 Voltage.- The heaters shall operate on both 115 volts direct current (d.c.) and single phase, 60 cycle, alternating current (a.c.), or other voltage, as specified (see 6.1).

3.4 Construction.- Heaters having a wattage of 75 watts per square foot or less shall be constructed with outer sheet metal panels of corrosion-resisting aluminum alloy 52S. The thickness of the front panel shall be at least 0.050 inch and the rear panel at least 0.032 inch. On heaters having a wattage density greater than 75 watts per square foot of heated panel area, the outer sheet metal panels shall be corrosion-resisting steel. The front panel shall be AISI type 430 with a minimum thickness of 0.025 inch and the rear panel shall be AISI type 302 with a minimum thickness of 0.019 inch.

3.4.1 Surface treatment.- The front face of aluminum panel heaters shall be painted with a high emissivity paint giving a dull or flat grey finish. The paint shall be heat resistant to the degree that it will not be discolored, damaged, or otherwise adversely affected by the heat produced. Corrosion-resisting steel panel heaters shall have the front face covered with flock for protection of personnel. The finished surface of the heater shall have a grey color. The back of the heater shall be left unpainted.

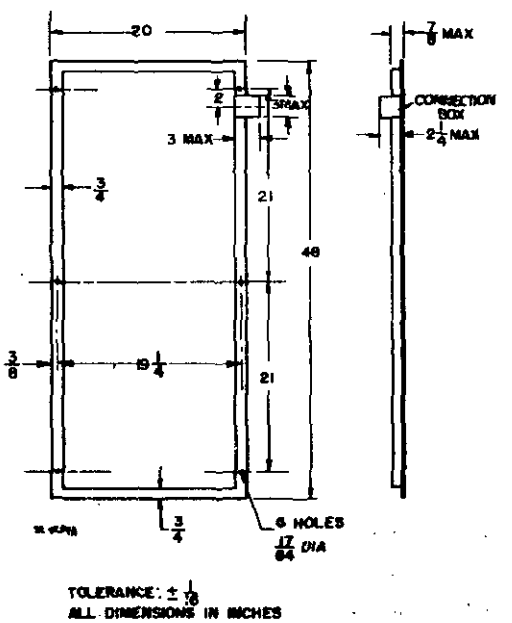


Figure 1 - Standard size panel heater.

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3.4.2 Enclosure. - The heaters shall be watertight in accordance with MIL-STD-108.

3.5 Mounting dimensions. - Each heater shall have a minimum of 6 mounting bolt holes 17/64 inch diameter located on the flanges. The mounting flanges shall be 3/4 inch wide. For standard heaters, the number and spacing of mounting bolt holes shall be as shown on figure 1.

3.6 Heating element. - The heating element shall be in sheet form arranged to provide a uniform heat dissipation to the panel surface. The heating element shall be insulated from ground with sheets of silicone bonded glass cloth.

3.7 Heat insulation. - The heating element shall be backed by a pad of glass wool to limit the amount of heat conducted to the rear surface.

3.8 Connection box. - Heaters shall be provided with a watertight connection box containing a terminal block. The connection box shall be located on the side of the panel and shall be removable to permit drilling by the installing activity to provide for the entrance of the supply cable.

3.9 Fittings. - Fittings for mounting shall be furnished with the heaters. These fittings shall consist of all bolts, washers, nuts and spacers necessary for installing the heater. Fittings for aluminum alloy panels shall be aluminum alloy or plated with cadmium or zinc.

3.10 Electrical requirements. -

3.10.1 Dielectric strength. - The heaters shall be designed and constructed so as to withstand for a period of 1 minute, a dielectric test voltage of twice the rated voltage plus 1000 volts (see 4.4.3).

3.10.2 Insulation resistance. - The insulation resistance of each heater when at normal operating temperature shall be not less than 10 megohms. The insulation resistance at normal room temperature shall be not less than 20 megohms (see 4.4.4).

3.10.3 Creepage and clearance. - Creepage and clearance distances shall be in accordance with MIL-E-917.

3.11 Temperature rise. - The maximum temperature rise at any point on the front of the aluminum panel heaters shall be 130° F. The maximum temperature rise on the front of the corrosion-resisting steel panel heaters shall be such that with the heater operating at normal room temperature (approx. 75° F.) a hand can be held firmly against the flock for 10 seconds without discomfort. The maximum temperature rise at any point on the rear panel of all heaters shall be 90° F.

3.12 Overtemperature protection. - The heater shall incorporate a means of protection against overheating by opening the circuit if a temperature of 400° to 475° F. is reached at any point on the front panel of the heater. The means of overtemperature protection shall be clearly illustrated on the drawing and shall require approval of the bureau or agency concerned.

3.13 Identification plates. - Each heater shall be provided with a corrosion-resisting metal or anodized aluminum identification plate with marking produced by etching, engraving, stamping, casting, or photoprocessing. The marking shall be plainly visible, preferably black in color against a light background. The plate shall be securely attached to the front of the heater. The plate shall contain the following information:

(a) Heater, panel.
(b) Voltage.
(c) Wattage.
(d) Manufacturer's name and type number.
(e) Contract or order number.
(f) Year of manufacture.

3.14 Drawings. - Drawings conforming to class A of MIL-D-963 shall be furnished. In addition to the information required by MIL-D-963, the drawings shall contain the following minimum data:

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(a) Schematic wiring diagram of the heater elements.
(b) Detail of portion of heating element to show method of construction.
(c) Detail of overtemperature protective means.
(d) Overall and mounting dimensions.
(e) Cross-section detail identifying each material.
(f) Detail of terminal connection.
(g) Detail of identification plate.

3.15 Workmanship.- Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.-

4.2.1 Inspection lot.- All heaters of the same size and wattage rating presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 Sampling for group A examination and tests.- Sample heaters shall be selected from each lot in accordance with MIL-STD-105, at inspection level III, with an acceptable quality level (A.Q.L.) of 1.5 percent defective. Each heater so selected shall be subjected to each of the group A examination and tests listed in 4.3.1. If any heater has any defect or fails in any test, it shall not be offered for delivery. If the number of defects exceeds the acceptance number the lot shall be subject to rejection.

4.2.3 Sampling for group B tests.- Sample heaters shall be selected from each lot in accordance with MIL-STD-105, at inspection level L-8 with an A.Q.L. of 4.0 percent defective. Each heater so selected shall be subjected to each of the group B tests listed in 4.3.2. If any heater fails in any test, it shall not be offered for delivery. If the number of defective heaters exceeds the acceptance number the lot shall be subject to rejection.

4.2.4 Sampling for group C tests.- Sample heaters shall be selected from each lot in accordance with MIL-STD-105, at inspection level L-3. Each heater so selected shall be subjected to each of the group C tests listed in 4.3.3. If any heater fails in any test, the lot represented shall be subject to rejection and the contractor shall correct the cause of failure in all equipment not yet shipped and make the same changes in all future production. Sample units shall be selected from the corrected lots and tested sufficiently to demonstrate the effectiveness of the changes.

4.3 Quality conformance inspection.-

4.3.1 Group A examination and tests.- Each of the sample heaters selected in accordance with 4.2.2 shall be examined and subjected to each of the tests specified herein, and the results of each test compared with specification requirements.

Group A examination and tests	Reference
Adjustment, fit and materials	4.4.1
Dimensions	4.4.2
Dielectric	4.4.3
Insulation resistance (cold)	4.4.4

4.3.2 Group B tests.- Each of the sample heaters selected in accordance with 4.2.3 shall be subjected to each of the tests specified herein, and the results of each test compared with specification requirements.

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Group B tests	Reference
Wattage	4.4.5
Insulation resistance (hot)	4.4.4

4.3.3 Group C tests.- Each of the sample heaters selected in accordance with 4.2.4 shall be subjected to each of the tests specified herein and the results of each test compared with specification requirements.

Group C tests	Reference
Temperature rise	4.4.6
Enclosure	4.4.7
Overtemperature protection (subsequent tests waived)	4.4.8

4.4 Examination and test procedures.-

4.4.1 Adjustment, fit and materials.- Each of the sample heaters shall be subject to a thorough examination to ascertain that the materials, workmanship and design are in strict conformance with the specification, contract and approved drawing. The weights of heaters shall be taken and recorded on the test report.

4.4.2 Dimensions.- Each of the sample heaters shall be measured to determine conformance with the applicable drawing.

4.4.3 Dielectric.- The test voltage shall be equal to twice the rated value of the terminal voltage plus 1,000; the frequency shall be not less than 60 cycles. The voltage wave shall approximate a sine wave. The testing voltage shall be applied continuously for a period not less than 60 seconds.

4.4.4 Insulation resistance.-

4.4.4.1 Each heater shall have insulation resistance measurements made to determine conformance with 3.10.2. Tests shall be made using a potential of 500 volts d.c.

4.4.4.2 For group A tests, the insulation resistance test shall be made at normal room temperature following the dielectric test.

4.4.4.3 For group B tests, the insulation resistance test shall be made immediately following the wattage test.

4.4.5 Wattage.- The watts input to the heater shall be measured at rated voltage at the end of one hour of operation and shall be within plus or minus 10 percent of the identification plate rating.

4.4.6 Temperature rise.- The temperature rise at the center of the aluminum panel heaters and 5 other points selected at random on the panel shall be measured to determine conformance with 3.11. Measurements shall be made on both the front and rear panels. This test shall be conducted with the heater in a vertical position, in still air and with no large heat absorbing or reflecting objects nearby. Temperatures shall be measured by means of thermocouples. To check the temperature rise of corrosion-resisting steel panel heaters for conformance with 3.11, the inspector shall hold his hand firmly against the flock.

4.4.7 Enclosure.- The heater enclosure shall be tested for watertightness in accordance with MIL-STD-108.

4.4.8 Overtemperature protection.- To test the effectiveness of the overtemperature protective means, a 6 inch square by 1 inch thick pad of glass wool shall be placed against the panel surface at a location selected at random. The panel surface temperature beneath the center of the glass wool pad shall be measured with a thermocouple. The circuit shall open at a temperature between 400° and 475°F.

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If the panel surface temperature beneath the center of the glass wool pad stabilizes before a temperature of 475°F. is reached, then the voltage shall be gradually increased until the circuit opens or until a temperature of 475°F. is reached. Opening of the circuit before a temperature of 475°F. is reached is required in order to comply with 3.12. This test shall be conducted a total of 5 times at random locations on the panel surface. After each test the heater shall be opened and repaired. Inasmuch as this test is destructive in nature, it may be conducted before the heater is sealed, so that the heater may be more easily repaired or restored to its original condition before it is offered for delivery. This test shall be conducted only on the first heater of each particular size and design offered by the manufacturer. Heaters of the same size and design need not be retested on subsequent contracts or orders; therefore, a record of the test certified by the Government inspector shall be retained by the manufacturer. This record shall be accepted on future orders as proof of conformance with 3.12.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements, and levels of preservation, packaging, packing and marking requirements for shipment shall be specified by the procuring activity (see 6.1).

(The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging.-

5.2.1.1.1 Level A.- Heaters shall be individually preserved and unit packaged in accordance with method III of MIL-P-116.

5.2.1.2 Packing.-

5.2.1.2.1 Level A.- Heaters shall be packed for overseas shipment in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class 2
PPP-B-640	Class 2

Box closure, strapping or banding shall be in accordance with the applicable box specification or appendix thereto.

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5.2.1.2.2 Level B.- Each heater shall be packed for domestic shipment and storage in containers conforming to one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Class 1
PPP-B-640	Class 1

Box closure shall be in accordance with the applicable box specification or appendix thereto.

5.2.1.2.3 Anchoring, blocking, bracing and cushioning of equipment shall be provided and shall be in accordance with JAN-P-100.

5.2.1.3 Marking.- In addition to any special marking required, interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and wattage rating if heaters are special size. If a specific number and spacing of mounting bolt holes is desired for special size heaters, this information shall also be specified (see 3.2 and 3.2.2).
(c) Special voltage other than that specified in 3.3.
(d) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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